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33. Optimal Page Replacement

Code:

```
#include <stdio.h>

int main() {    int pages[20],
frames[10];    int n, f, faults
= 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(int i = 0; i < n; i++)        scanf("%d",
&pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(int i = 0; i < f; i++)
frames[i] = -1;

    for(int i = 0; i < n; i++)
    {
        int found = 0;

        for(int j = 0; j < f; j++)
if(frames[j] == pages[i])
found = 1;

        if(!found)        {            int
pos = -1, farthest = -1;

            for(int j = 0; j < f; j++)
            {
if(frames[j] == -1)
```

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        pos = j;
break;
    }
    int k;        for(k = i +
1; k < n; k++)
if(pages[k] == frames[j])
break;
    if(k >
farthest)
    {
farthest = k;
pos = j;
    }
}

frames[pos] = pages[i];
faults++;
}
}

printf("Total Page Faults = %d\n", faults);

return 0;
} Output:

```

```

Enter number of pages: 8
Enter page reference string:
1 2 3 4 4 3 2 1
Enter number of frames: 3
Total Page Faults = 5

...Program finished with exit code 0
Press ENTER to exit console.

```